

Exponential Growth & Decay

Numberbender | WORKSHEET



Name: _____ Date: _____ Score: _____

Solve each differential equation and application problem. Show all work.

Calculus I Worksheet # 84

1	If $\frac{dN}{dt} = \frac{1}{5} N$, and $N(0) = 500$, then $N(t) =$
2	The rate of increase of a population is given by $\frac{dN}{dt} = \frac{1}{2} N$. If the initial population is 1000, then find the equation that will give the population at any time t .
3	The rate of mass decay for a certain radioactive substance is given by $\frac{dM}{dt} = -\frac{M}{50}$, where M is in grams and t is in years. If the initial mass of the substance is 250 grams, what mass will remain when $t = 100 \ln 2$ years?
4	Consider the curve $y^2 = 4 + x$ and the chord AB joining points A(-4,0) and B(0,2) on the curve. a) Find the x and y coordinate of the point on the curve where the tangent line is parallel to the chord AB. b) Find the area of the region R enclosed by the curve and the chord AB. c) Find the volume of the solid generated when the region R, defined in part (b), is revolved about the x -axis.
5	Find the first point of intersection of the curves $f(x) = \sin x$ and $g(x) = e^{-x}$.
	The graph is for Questions 6 through 8 . It shows the velocity, in ft/sec for $0 < t < 8$, of an object moving along a straight line.
6	Find the object's average speed (in ft/sec) for this eight second interval.
7	When did the object return to the position it occupied at $t = 2$?
8	Find the object's average acceleration (in ft/sec^2) for this 8-second interval.
9	$f(t) = \int_0^t \frac{1}{1+x^2} dx$, find $f'(t)$.



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Answer key — for instructor use only.

10	Find the maximum value of the function $f(x) = x^4 - 4x^3 + 6$ on the interval $[1,4]$.
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Answers:

1) $500e^{\frac{t}{5}}$	2) $N = 1000e^{\frac{t}{2}}$	3) 62.5 g	4a) (-3, 1) b) $\frac{4}{3} \approx 1.333$ c) $\frac{8\pi}{3} \approx 8.378$	5) $x = 0.589$
6) 1	7) never	8) $-\frac{1}{4}$	9) $\frac{2t}{1+t^4}$	10) 6

